

# From AI Copilot to Commander: Masterclass on agentic investigation

Discord: <https://discord.gg/fjeRgYha9>



BSIDES CANBERRA IX 2026



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Sr. Soln. Architect



Leo Meyerovich  
Founder & CEO  
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instructions @

[discord.com/invite/FTb0mrcm](https://discord.com/invite/FTb0mrcm)

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# Instructors



Sindre Breda  
Sr. Soln. Architect

Former Chief Inspector (Kripas),  
Senior analyst & engineer (Defendable)



Leo Meyerovich  
Founder & CEO  
@LMeyerov

Graphistry, Louie.ai, GFQL,  
Apache Arrow, Nvidia RAPIDS



AI alert RPE winner



First AI agent speed run

+ Thomas, Tanmoy, Dani



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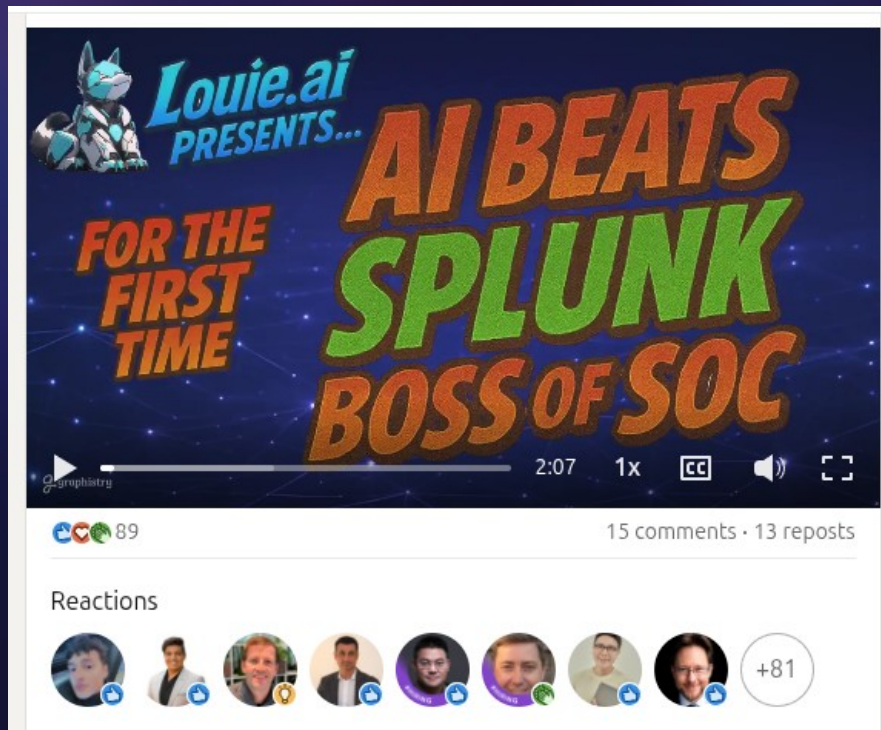
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[info@graphistry.com](mailto:info@graphistry.com)

# March 2025

First AI agent to autonomously pass: 70%+

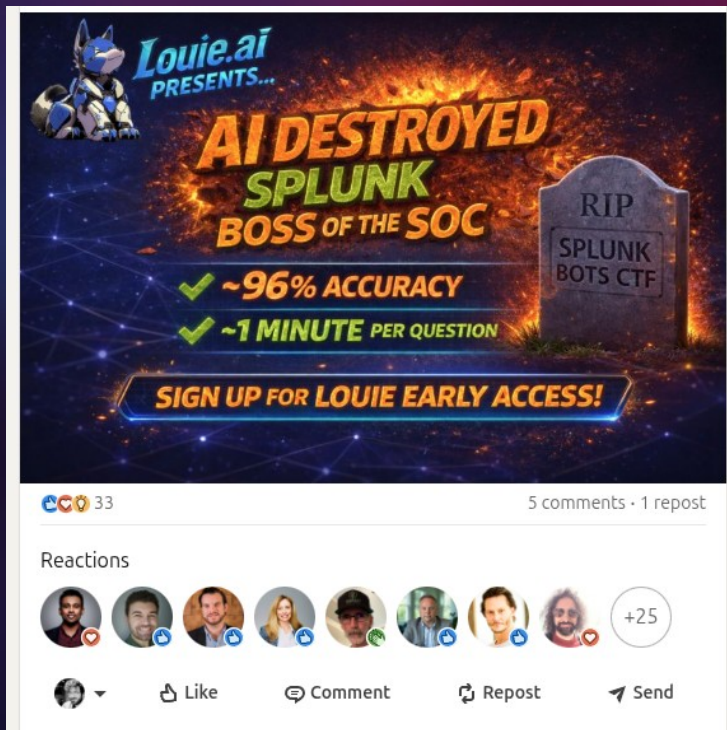


[https://www.linkedin.com/posts/leo-meyerovich-09649219\\_genai-activity-7324060032436731905-pC35/](https://www.linkedin.com/posts/leo-meyerovich-09649219_genai-activity-7324060032436731905-pC35/)

# March 2026

96%+

see also:  
[botsbench.com](https://botsbench.com)



[https://www.linkedin.com/posts/leo-meyerovich-09649219\\_the-main-new-s-is-the-louieai-team-is-declaring-activity-7440558663569723392-JJk4/](https://www.linkedin.com/posts/leo-meyerovich-09649219_the-main-new-s-is-the-louieai-team-is-declaring-activity-7440558663569723392-JJk4/)



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# Let's meet one another!

Poll:

1. Team (SOC, IR, TI, hunt, ...) + comfort in coding + comfort in vibe coding
2. Use case & learning goal



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[info@graphistry.com](mailto:info@graphistry.com)

# Goals

## Beginner

Learn code agents, floor + ceiling

## Intermediate

Shore up basics & identify big opportunities

## Advanced

Move towards evals, AI investigation factory, and AI SOC patterns

—

## Big picture

Concepts for genAI uses, implementation, and, industry trajectory, & team growth

## Practical

How to start or advance individually & as a team

## Use cases



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# Plan - 7hr (10:00 - 17:00)

**10:00-10:30** Intro

**10:30-11:15** Lab 1: Setup, first vibes

**11:15-12:00** AI; Harnesses; MCPs & semantic layer

**13:00-14:00** Lab 2a: BOTSv3 incident + vibes evals

**12:30-13:30:** LUNCH

**13:30-14:00** Lab 2b: BOTSv3 incident + vibes evals

**14:00-14:45:** CTFs & botsbench; Skills & planning

**14:45-15:00:** BREAK

**15:00-16:00:** Lab 3: Timelining with skills, vibes evals

**16:00 - 16:30:** Maturity - Evals & AI Investigation Factories

**16:30 - 17:00:** Group discussion



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# Plan - 2 day

## DAY 1

### Topics

About, **use cases**,  
agents (arch, MCPs, skills, planning),  
**evals**

### Interactive

Intros + EOD discussion

### Labs

Lab 1 - DIY LLM

Lab 2 - MCP, BOTS I, **exploring, evals I**

## Day 2:

### Topics

AI foundations, **decision framework**,  
**RAG & Memory**, planning II, **security & GRC**

AI investigation factory

### Interactive

Morning review + EOD discussion

### Labs

Lab 3 - Skills, **API, evals II**

Lab 4 - Timelining, **DIY**



# AI Copilot to Commander: You are in the right

Rec: Join for  
morning setup

Day alternates  
~1hr talk vs lab

**1. Discord:** <https://discord.com/invite/FTb9mrecm>

- Get creds, download links, slide links, etc

**2. Github:** <https://github.com/graphistry/bsides-canberra26-copilot>

- Slides will be provided on github end of training.
- Schedule, incl. Breaks!

**3. Install:**

- **Option A: BYO Coding Harness/LLM:**  
Splunk MCP creds via Discord; [hub.graphistry.com](https://hub.graphistry.com) (optional gpu graph viz)
- **Option B: Louie Investigation Harness** (download.louie.ai, free)  
Splunk MCP creds + OpenAI key via Discord; [hub.graphistry.com](https://hub.graphistry.com) (optional gpu graph viz)
- **Option C: ... + Python:**  
Do Option B + <https://louie-py.readthedocs.io> in Jupyter/Colab



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# Lab 1: Setup, viz, ctf q1 - 45 min

Github: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



# Pick your weapon

## BYO Harness & LLM

Setup Splunk MCP + Graphistry Hub creds & skills; BYO LLM

## Louie Desktop (free) + OpenAI (provided)

Setup Splunk MCP + Graphistry Hub creds & skills; OpenAI token (w/ Luna)

... + Python

... + louie-py client, esp via Jupyter



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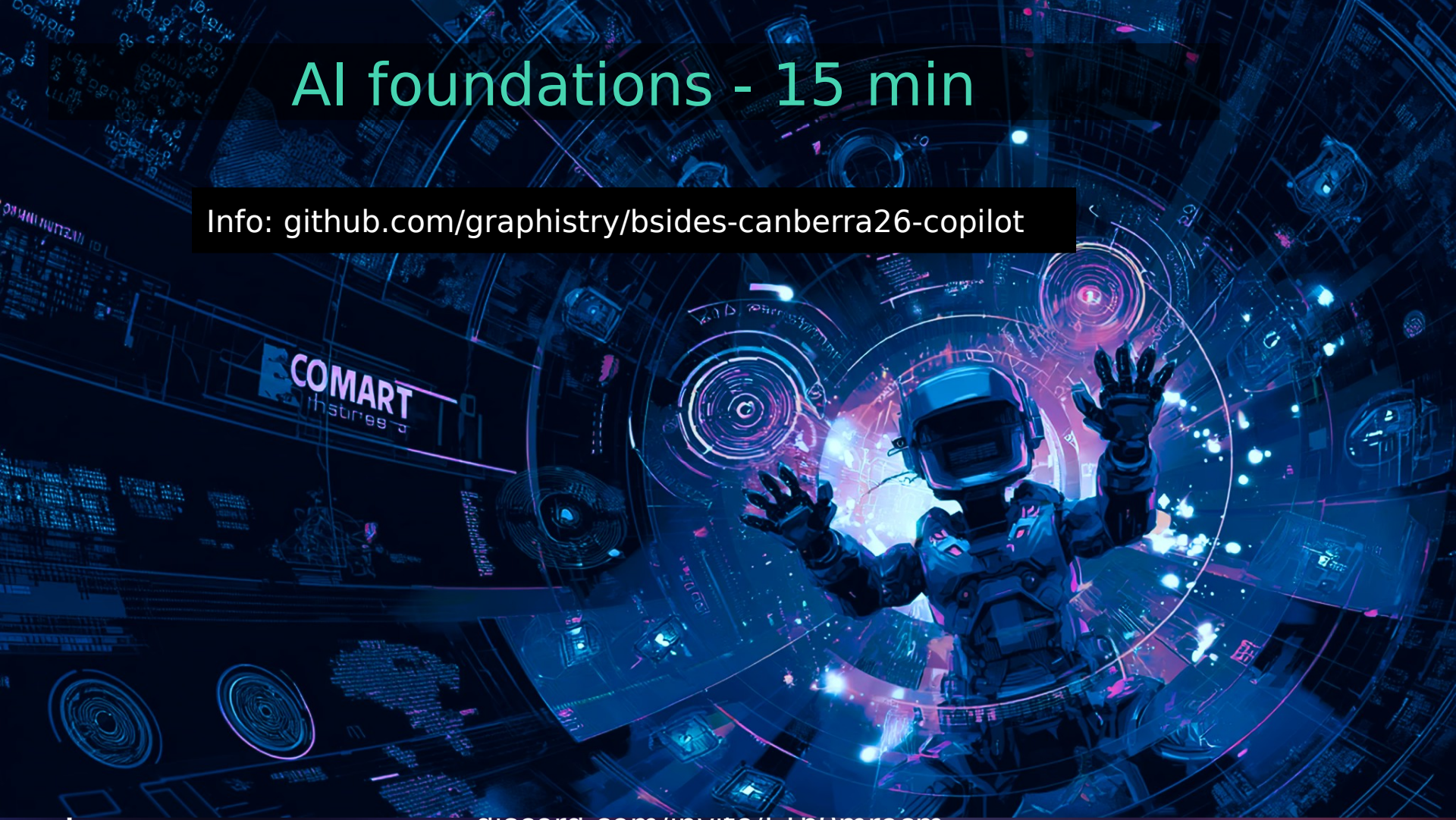
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# AI foundations - 15 min

Info: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



# AI thinks in vectors: Text2number with tokens

Tune to code, ...

1

| Tokens | Characters |
|--------|------------|
| 35     | 106        |

```
<Data Name='ParentCommandLine'>C:\Windows\system32\cmd.exe /c netstat -nao  
| findstr /r "LISTENING"</Data>
```

Text Token IDs

| Tokens | Characters |
|--------|------------|
| 35     | 106        |

[66361, 7317, 2053, 10510, 5266, 3665, 13376, 34, 16008, 20901, 59, 17360, 1398, 59, 18804, 39494, 820, 66, 2776, 12164, 533, 1503, 78, 1022, 1646, 689, 820, 81, 392, 44615, 1262, 2694, 55059, 1186, 29]

Text Token IDs

<https://platform.openai.com/tokenizer>

2

Token dictionary:

- 66361 => [0.123, 0.31, ...]
- 7317 => [0.0, 0.99, ...]
- ...

3

Combine chunk into 1 vector

- Sum, Max, Attention, ...

[0.123, 0.99, ...]



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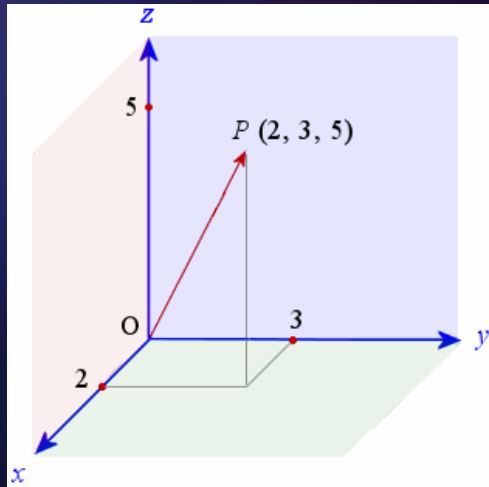
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# AI vectors live in high-dimensional spaces that are semantic

3-dim vector: [2.0, 3.0, 5.0]

OpenAI text-embedding-3-small: 1536 dimensions



```
C:\Windows\system32\cmd.exe /c netstat -nao |  
findstr /r "LISTENING"
```

```
netstat -nao
```

```
C:\Windows\system32\cmd.exe /c wmic os get  
LocalDateTime /value 2&gt;nul
```

```
"/bin/sh" "-c"
```

```
"/opt/splunkforwarder/etc/apps/Splunk_TA_nix/bin/  
ps.sh"
```



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# Misinfo embedding demo

<https://hub.graphistry.com/graph/graph.html?dataset=6fbdc5fb9ca64f37ade8a7a5ccb337f0&type=arrow&viztoken=085d0350-266f-4cd3-835a-43c08f5602a4&usertag=8c8dff2b-pygraphistry-0.11.7&splashAfter=false&info=true&play=0strongGravity=True&pointSize=0.3>



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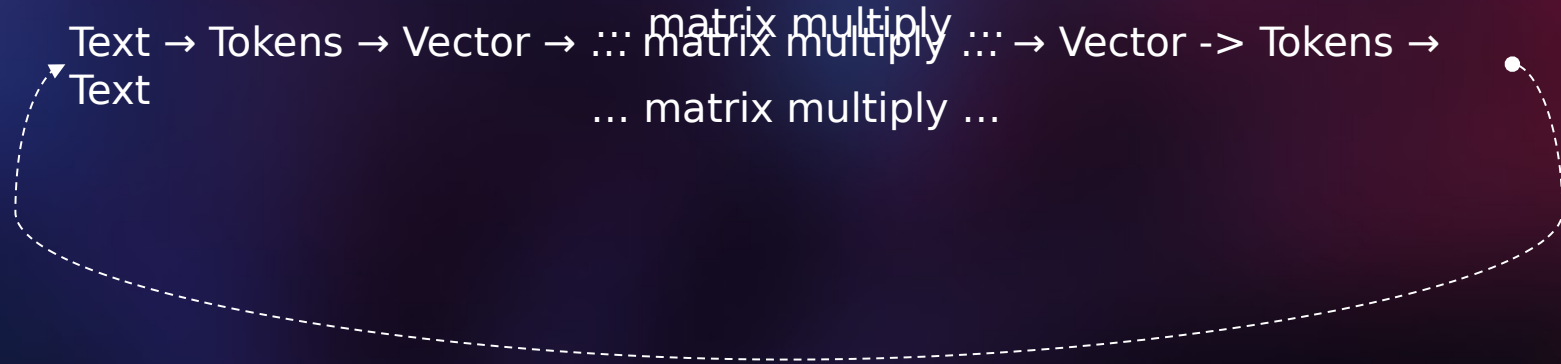
info@graphistry.com



# Do not think about strawberries

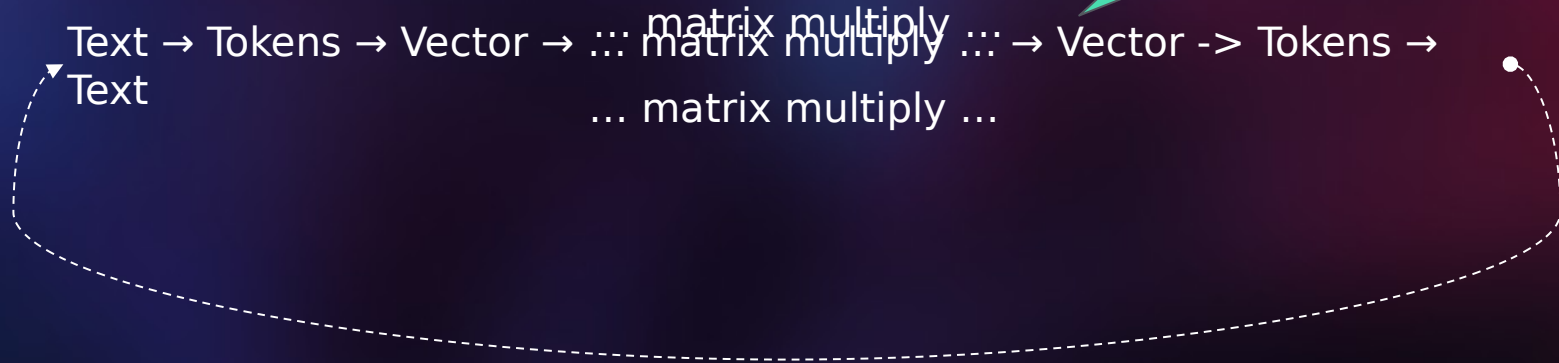
LLMs are leaky abstractions

# Generative AI is an *autoregression* loop

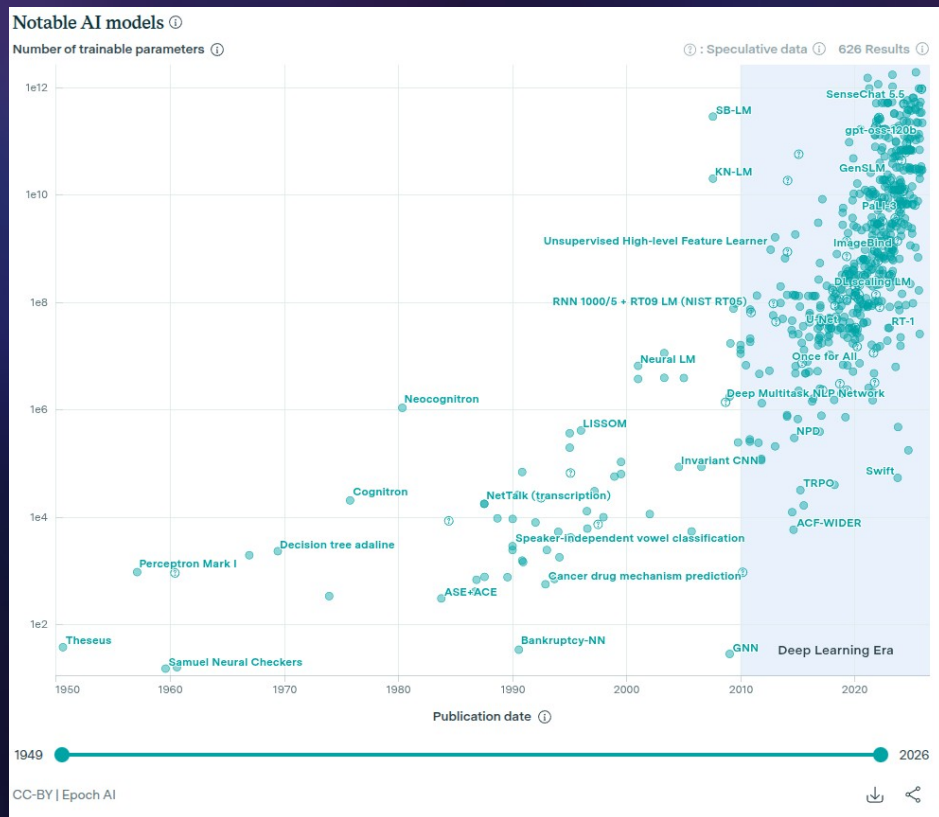


# Generative AI is an *autoregression* loop

Classifier (Jev):  
Return only ~1 token



# AI is just matrix multiply engines: Sizing



**Model size is # of matrix weights**

**Scaled 1,000X then hit a wall**

- BERT (2018): 340M
- GPT-2 (2019): 1.5B
- GPT-3 (2020): 175B
- GPT-4 (2023): 1.7T (MoE)

**Mixture of Experts: Memory reuse**

Ex: GLM 5.2 on 4-8 x H200 GPU's

Ex: DBRX on 4 x H200 GPU

- 75% weights, 25% KV cache
- 132B total params trained
- Only 36B active per token
- ... 4 experts per token



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# Sparse MoE (Mixture of Expert) & Quantization:

## GLM 5.3 Case Study: Base vs Flash, Quant vs REAP

### GLM 5.3: Post-training of GLM 5.2

**8 x H200:** 744B (40B active) weights  
+ 12-50GB KV Cache

- Do not “activate” all weights:  
less bandwidth & compute
- On each layer (78 layers):  
1 token -> 8 experts (of 256) + 1 shared
- Stored as FP8; active as BF16

### GLM 5.3 W4AFP8 REAP (Aikido)

**4 x H200**

- 753B (40B active) at int4
- REAP experts 256 -> 168

### GLM 5.3 Flash: New base model

**4 x H200:** 321B (188B active)

- 45 layers:  
1 token -> 8 experts (of 288) + 1 shared
- FP8 and BF16 checkpoints

### GLM 5.3 Flash W4AFP8 (Graphistry):

**2 x H200**

- 321B (18B active) at int4
- Activations: fp8



OPEN WEIGHTS PROPRIETARY OUTLINED: GLM 5.3 RUNS.

|                                                                 |                        |       |        |                 |
|-----------------------------------------------------------------|------------------------|-------|--------|-----------------|
| GPT-5.6 Sol (medium)<br>Louie.ai                                | <div><div></div></div> | 45/59 | 3m 16s | \$1.77/solved   |
| Opus 4.8<br>opencode 1.16.2                                     | <div><div></div></div> | 40/59 | 58s    | \$0.835/solved  |
| <b>GLM 5.3 Flash</b><br>Louie.ai                                | <div><div></div></div> | 38/59 | 4m 58s | \$0.125/solved  |
| GLM 5.2<br>Louie.ai                                             | <div><div></div></div> | 37/59 | 3m 15s | \$0.764/solved  |
| <b>GLM 5.3 Flash (W4AFP8)</b><br>Louie.ai (self-hosted, 2xH200) | <div><div></div></div> | 35/59 | 2m 44s | ~\$0.348/solved |
| Opus 4.5<br>Louie.ai                                            | <div><div></div></div> | 35/59 | 3m 53s | \$1.36/solved   |
| GPT-5.6 Luna (high)<br>Codex CLI                                | <div><div></div></div> | 34/59 | 66s    | ~\$0.062/solved |
| GPT-5.6 Terra (medium)<br>Louie.ai                              | <div><div></div></div> | 34/59 | 1m 47s | \$1.10/solved   |
| Qwen3.7 Plus<br>opencode 1.16.2                                 | <div><div></div></div> | 33/59 | 24s    | \$0.158/solved  |
| GLM 5.2<br>opencode 1.16.2                                      | <div><div></div></div> | 33/59 | 1m 49s | \$0.595/solved  |
| Kimi-K3<br>opencode 1.16.2                                      | <div><div></div></div> | 33/59 | 2m 02s | \$0.844/solved  |
| Opus 5 (high)<br>Claude Code                                    | <div><div></div></div> | 33/59 | 77s    | \$1.15/solved   |
| <b>GLM 5.3 Flash</b><br>opencode 1.18.25                        | <div><div></div></div> | 32/59 | 1m 53s | \$0.082/solved  |
| GPT-5.6 Terra (medium)<br>Codex CLI                             | <div><div></div></div> | 32/59 | 53s    | ~\$0.678/solved |
| GPT-5.6 Luna (high)<br>Louie.ai                                 | <div><div></div></div> | 31/59 | 3m 04s | \$0.638/solved  |
| Sonnet 5 (medium)<br>Claude Code                                | <div><div></div></div> | 30/59 | 76s    | \$1.20/solved   |
| <b>GLM 5.3</b>                                                  | <div><div></div></div> | 29/59 | 2m 05s | \$1.26/solved   |

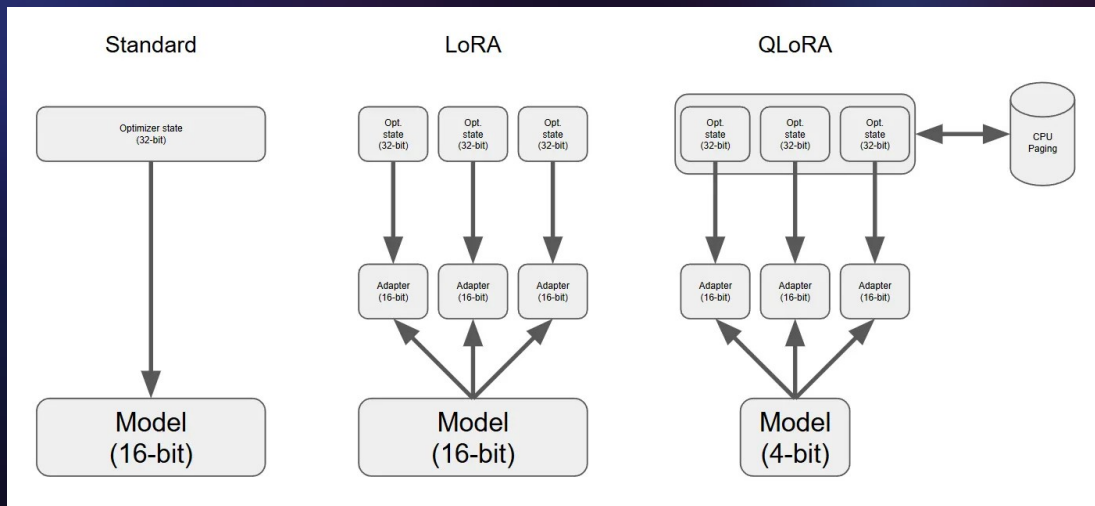
# Fine-tuning: QLoRA - Skinnier base models w/ per-layer patches

**Fine-tuning:** Retrain on new examples for your use case

**Problem:** Expensive to retrain full 1T weight model...

**Solution:** High-precision patch matrix per network layer + lower precision base model

second life in agent era?



float16 patches: 1% of model

4x smaller base: float16->nvfp4



# Modern harnesses, thick MCPs - 20 min

Info: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



# Agentic AI is a tool-calling chat loop between actors

<user>  
find the bug  
</user>

<system>  
ok  
</system>

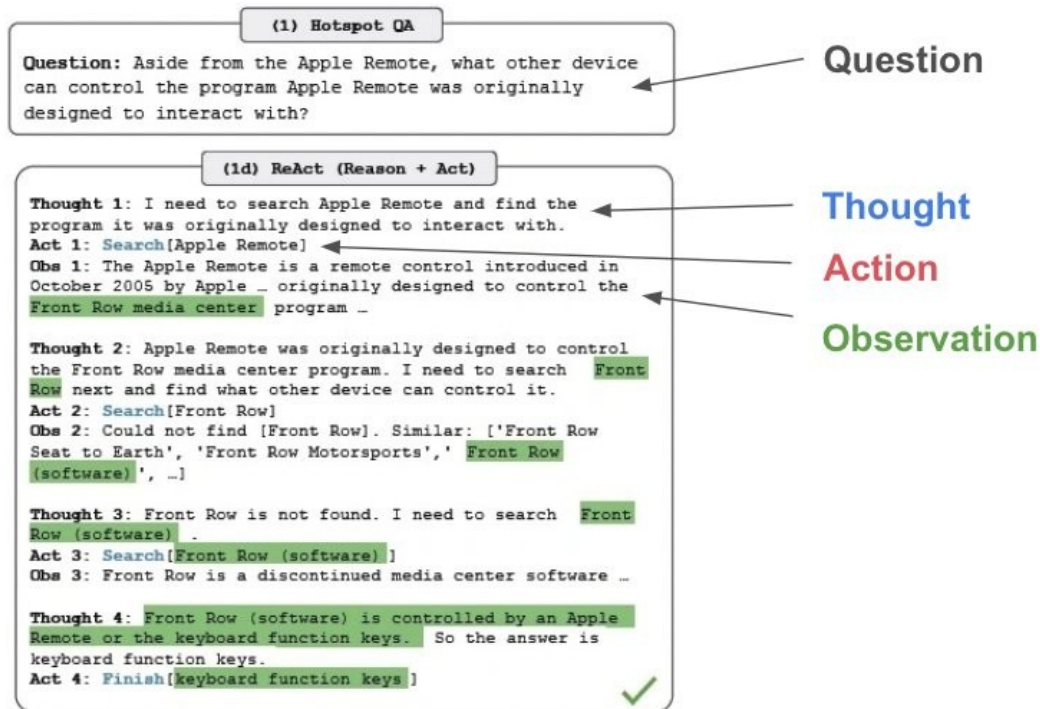
<tool\_call id="lint\_analyzer">  
  <param>...</param>...  
</tool\_call>

...

*Trace / trajectory of turns*



# ReAct Agent



From “*REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS*” (Shunyu et al., 2022)

PROMPT\_TEMPLATE = <<~PROMPT

Today is %{today} and you can use tools to get new information. Answer the question as best as you can using the following tools:

%{tool\_description}

Use the following format:

Tool call

**Question:** the input question you must answer

**Thought:** comment on what you want to do next

**Action:** the action to take, exactly one element of [%{tool\_names}]

**Action Input:** the input to the action

**Observation:** the result of the action

... (this Thought/Action/Action Input/Observation repeats N times, use it until you are sure of the answer)

Thought: I now know the **final answer**

Final Answer: your final answer to the original input question

Begin!

Question: %{question}

Thought: %{previous\_responses}

PROMPT

[ReAct Agent in 150 lines of code](#)

## Reasoning models: What if the model internalized CoT? ← Faster, smarter

# USER

```
----  
input
```

“effort”

LLM

```
out1 ---> think2 ---> think3 ---> out4 --> think5 --> out6 --> out7
|         |         |         |         |         |
v         v         v         v         v         v
```

## AGENT HARNESS

```
out1 ---- [summary think2-3] ---> out4 -----> out6 --> out7
          (lossy)
```

## CHAT UI

```
out1 ----- out4 -----> out6 --> out7
```

# Reasoning models: What if the model internalized CoT? ← Faster, smarter

USER

-----

input ----->

LLM

---

out1 ---> think2 ---> think3 ---> out4 --> think5 --> out6 --> out7  
| | | | | | |  
v v v v v v v

Unsafe!

AGENT HARNESS

-----

out1 ---- [summary think2-3] ---> out4 -----> out6 --> out7  
                    (lossy)

Nicer UX!

CHAT UI

-----

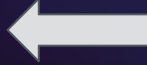
out1 ----- out4 -----> out6 --> out7

limit stealing  
(distillation)



# X-Ray vision: Agent harness edits the context each turn

```
""  
{# highly stable #}  
{ system_prompt }  
{ safety_policies }  
{ output_contract }  
{ user_profile }  
{ available_tools }  
{ available_skills }
```



```
{# medium stability #}  
{ active_tools }  
{ active_skills }  
{ long_term_memory }  
{ goals_loops }
```

```
{# volatile #}  
{ retrieved_context }  
{ history }  
{ inbox }  
{ current_state }  
{ user_input }  
""
```

  
agent loop

- Orchestrator
- System prompt
- User profile
- Sessions
- Identity
- Memory
- Hooks
- Tools
- Skills
- Plans
- Goals
- Subagents
- Reflection
- Policy
- Models
- Artifacts
- Telemetry

## Learning & usage

### tip:

Many tools let you  
see the context  
window or at least  
inner turns

... keeps growing ...



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# X-Ray vision: Agent harness edits the context each turn

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{ long_term_memory }  
{ goals_loops }
```

```
{# volatile #}  
{ retrieved_context }  
{ history }  
{ inbox }  
{ current_state }  
{ user_input }  
""
```

  
agent loop

- Orchestrator
- System prompt
- User profile
- Sessions
- Identity
- Memory
- Hooks
- Tools
- Skills
- Plans
- Goals
- Subagents
- Reflection
- Policy
- Models
- Artifacts
- Telemetry

## Learning & usage tip:

Many tools let you see the context window or at least inner turns

User-controllable structured programming

External harness  
cases  
environment

...

... keeps growing ...



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## DEPLOYMENT

local vs remote



## AUTONOMY

interactive vs automated vs



## STATE

id, durable, migratable, self-updating



```
+-----+
| POLICY / CONTROL |
| auth • perms • audit |
+-----+
```

↓  
v

```
+-----+ +-----+ +-----+ +-----+
| User / | --> | Agent | --> | Tool Calls | --> | State / Resources |
+-----+ +-----+ +-----+ +-----+
|                                     |
| ^                                  |
| |                                  |
| |                                  |
| |                                  |
| |                                  |
+-----+ +-----+ +-----+ +-----+
| Data (SIEM/DB) |
| Memory / Retrieval |
| Artifacts |
| Execution (sandbox) |
| External APIs |
```



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# Tools: Different protocols solve different problems

## **MCP:** Self-describing tools & resources

Use for secure, discoverable remote capabilities

```
@tool(name=...) def foo(): ...
```

## **skills.md:** Resource folders including markdown

Use for workflows, knowledge, programs

Use `gh` to create/edit/monitor PRs

## **A2A:** Self-describing agents (Google)

Use for agents not just APIs; too early / MCP absorbs?

```
@agent(name="SOC_Investigator", capabilities=["analyze_logs",  
"correlate_threats", "generate_reports"]) class SOCAgent: ...
```



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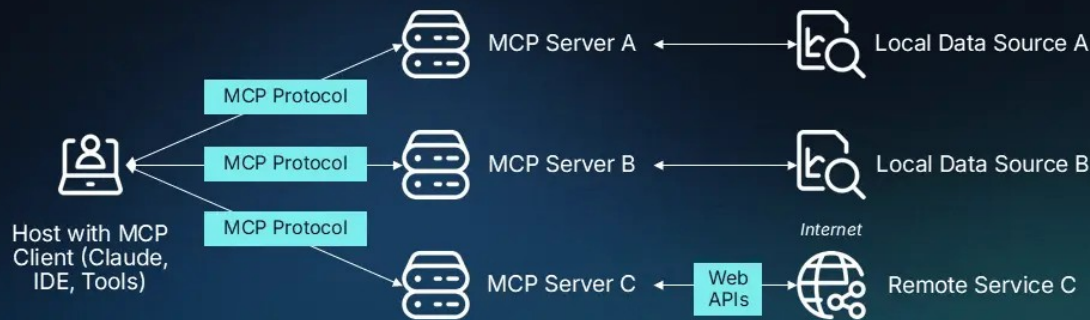
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# MCP 1/3: The Bluetooth for AI Agents - Wire protocol

<https://modelcontextprotocol.io/>

## MCP Architecture

descope



- Akin to OpenAPI + resources, natural language
- Main case: Vendor-provided remote API

**LLM Semantics:** Resources that agents can discover and load into context securely + efficiently

**MCP Host:** Ex - Claude Code

**MCP Server:** Capabilities

Local: Host runs code ← CLI/Py better?

Remote: Service provider runs for you

**MCP Client:** Host's conn to server

**Transport:**

- stdio - CLI
- HTTP streaming - web

**Elements**

Tools, Resources (src, img, ..), Prompts

**Payloads**

- Text
- JSON

[info@graphistry.com](mailto:info@graphistry.com)

instructions @

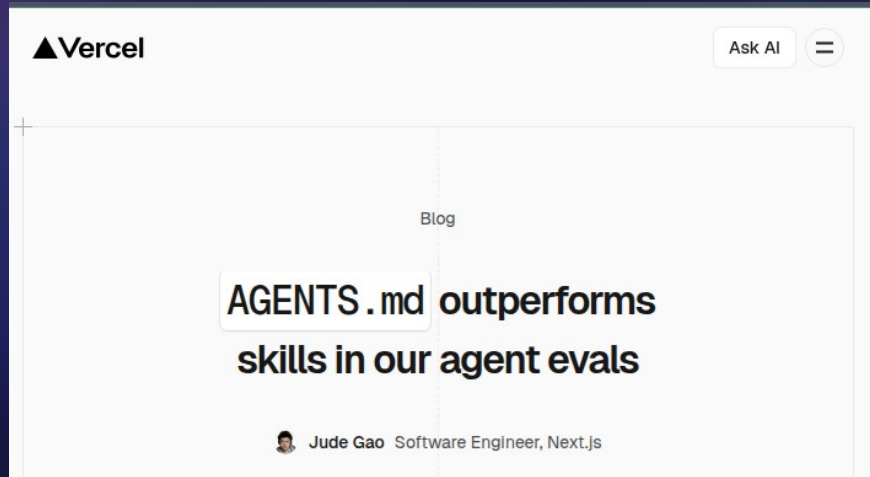
[discord.com/invite/FTb0mrcm](https://discord.com/invite/FTb0mrcm)

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# MCP 2/3: Context engineering



Viral Vercel post w/ great context engineering...  
but blamed MCP vs their misunderstanding

Vercel's trick: Put Table of Contents in AGENTS.md

Problem: Wins their eval battle, loses war on bloat

## 1. Problem

AI does not know how to use MCP tools

## Context Engineering

Include docs, especially examples, on tools

## 2. Problem

Harness bloats context with per-tool docs

## Context Engineering

### User

Ensure MCPs lazily / progressively loaded

### Developer

Make prompt resource TableOfContext.md  
-> prompt resources Chapter<XYZ>.md



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# MCP 3/3: Security

## Authentication

Env (local), HTTP Headers, OAuth 2.0

## Authorization

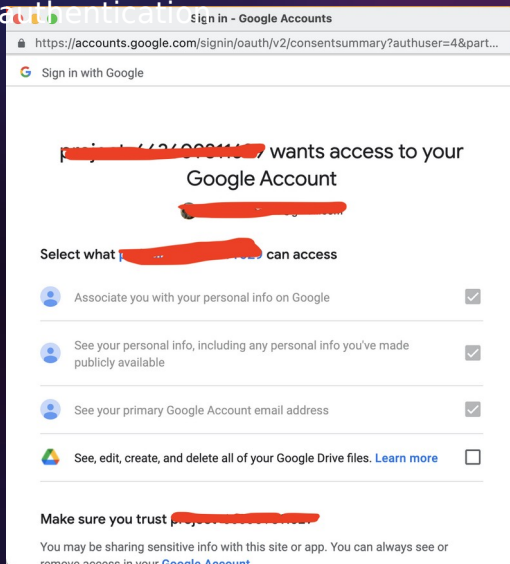
stdio: nothing; HTTP: OAuth scopes

Practice trending to more dynamic:

- OAuth roles ← sad UX
- MCP host toggles
- AI auto decision mode

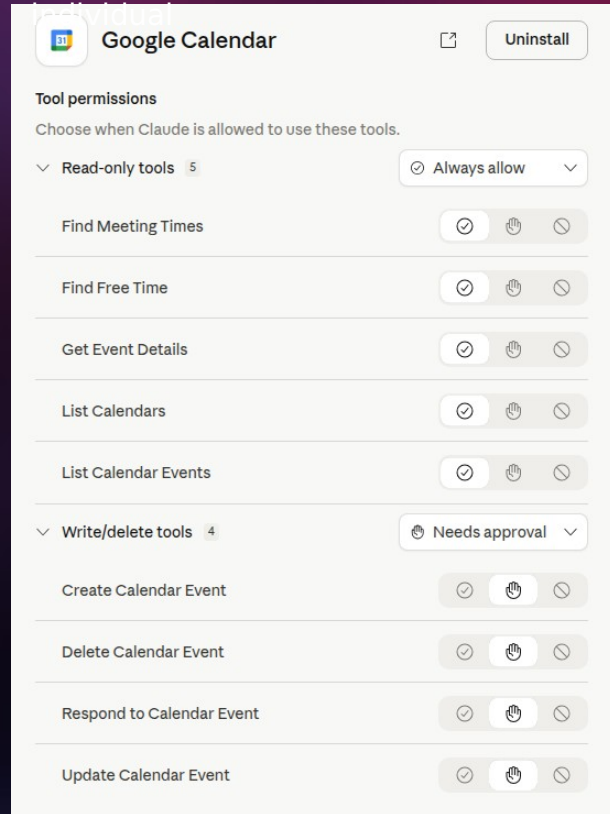
Day 2: More options

Option - OAuth 2.0 scopes:  
Authorizations tied to authentication

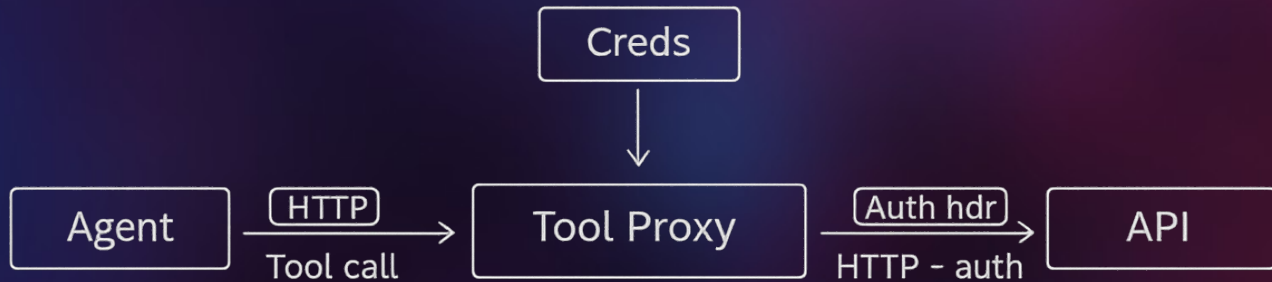


<https://stackoverflow.com/questions/70765918/google-oauth-consent-screen-how-do-i-know-the-user-grants-the-requested-access>

Option - MCP Host toggles:  
Dynamic toggles: read/write &



# Alt: Secretless tool calls with credential broker injection proxy



Challenge - Policy: Credentials != authorization

Challenge - Integration: Harness-native vs external



# Ex: Databases - Thin wrapper vs semantic layer

"search sourcetype=cisco:aws host=172.16.24.24"

What does field "host" mean?

Logging device? Email provider? 90's Talk show host turned podcaster?

Are all the fields even indexed extractions?

Syntax/type/convention errors? Perf violations?

evals  
target

**Semantic layer:** Symbolically-extracted + AI-enriched

- **Virtual fields:** Unparsed log fields
- **Foreign key relationships:** Pivot on IP -> User -> ...
- **Multi-fields, top queries, semantic ambiguity, ...**

→ **More good:** Discover where & how to look

→ **Less bad:** Invalidate & explain bad syntax, schema, style



GRAPHISTR



Louie.ai

instructions @

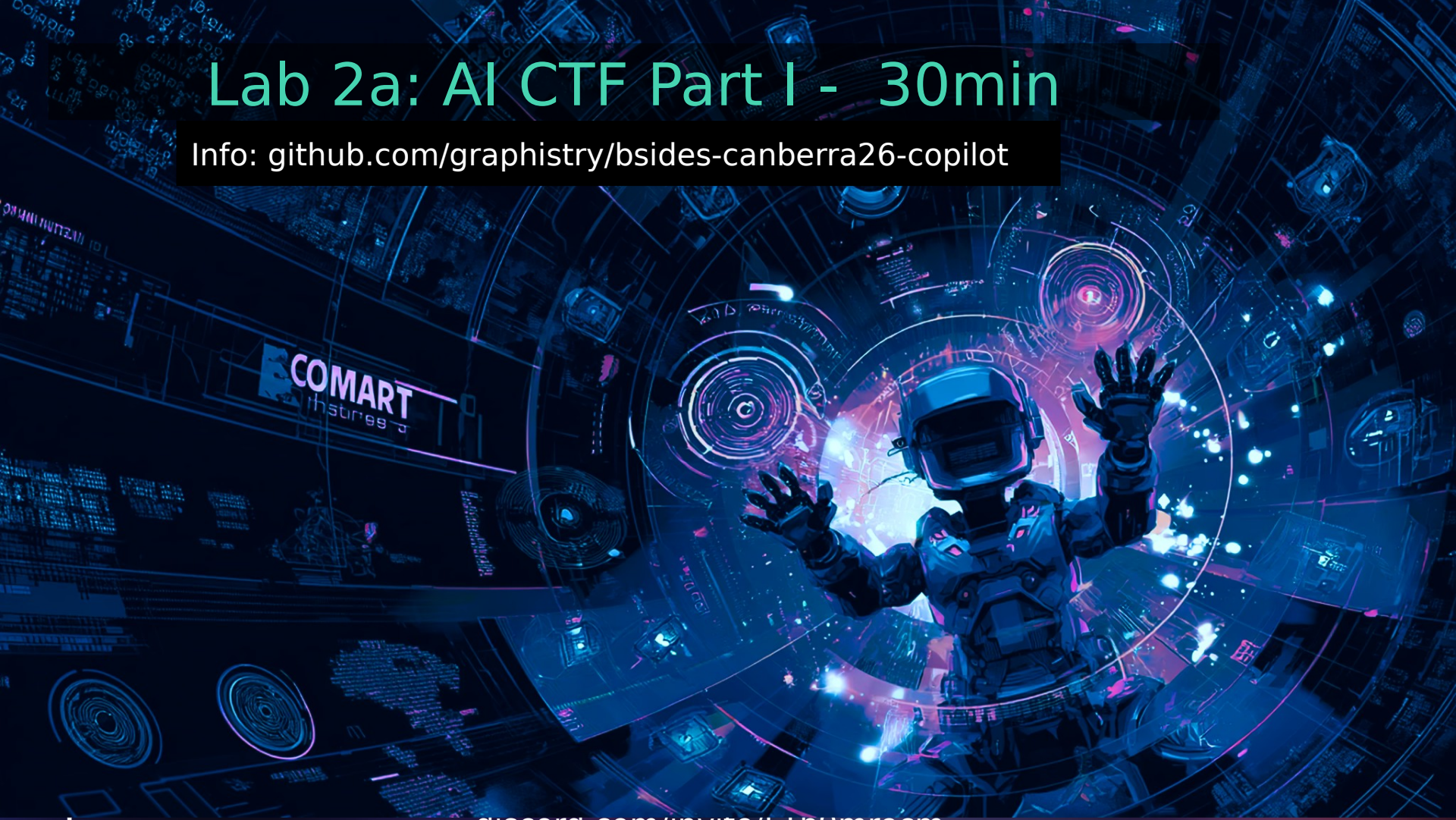
discord.com/invite/FTbQmrcm

phistry.com

35

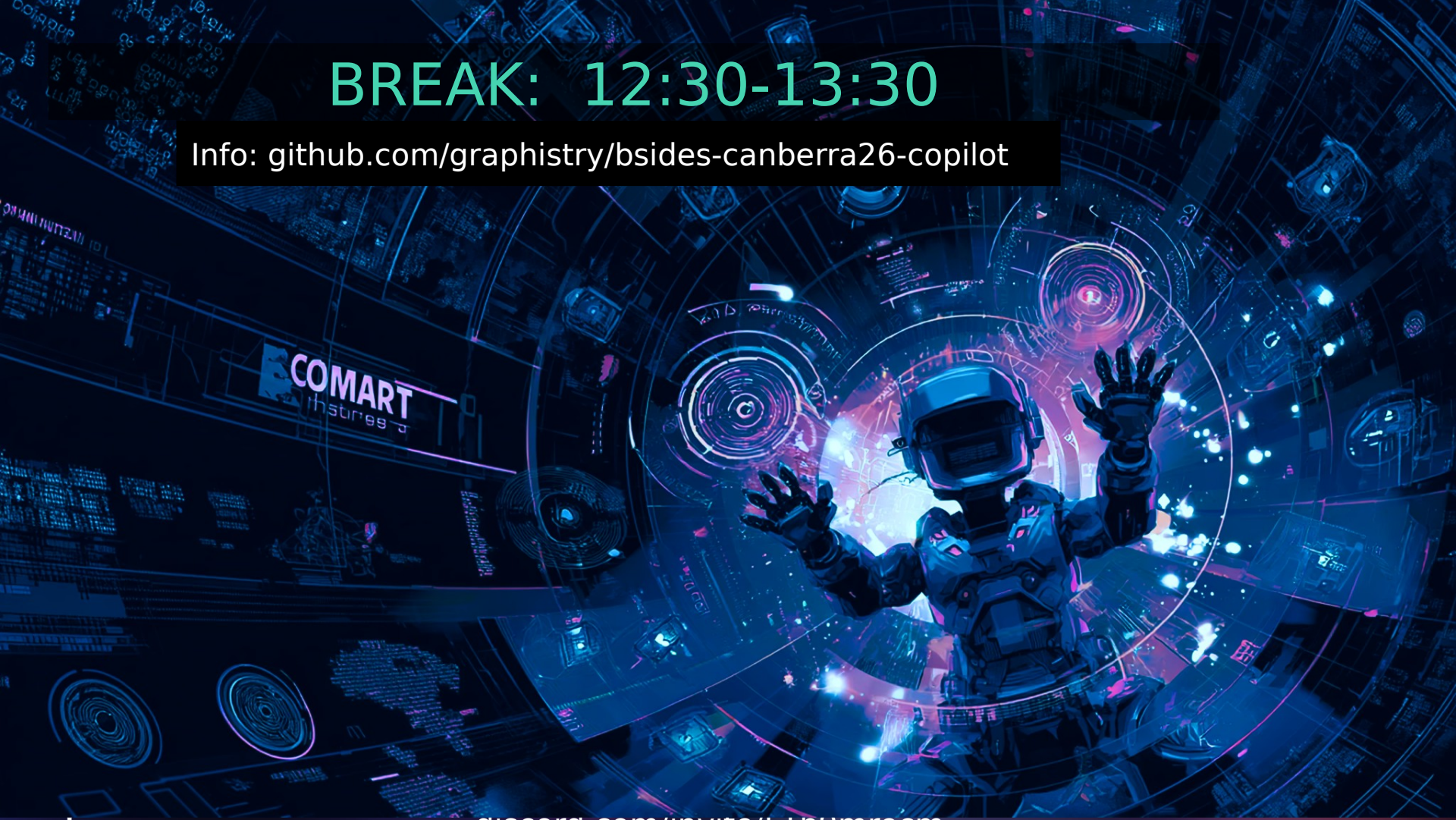
# Lab 2a: AI CTF Part I - 30min

Info: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



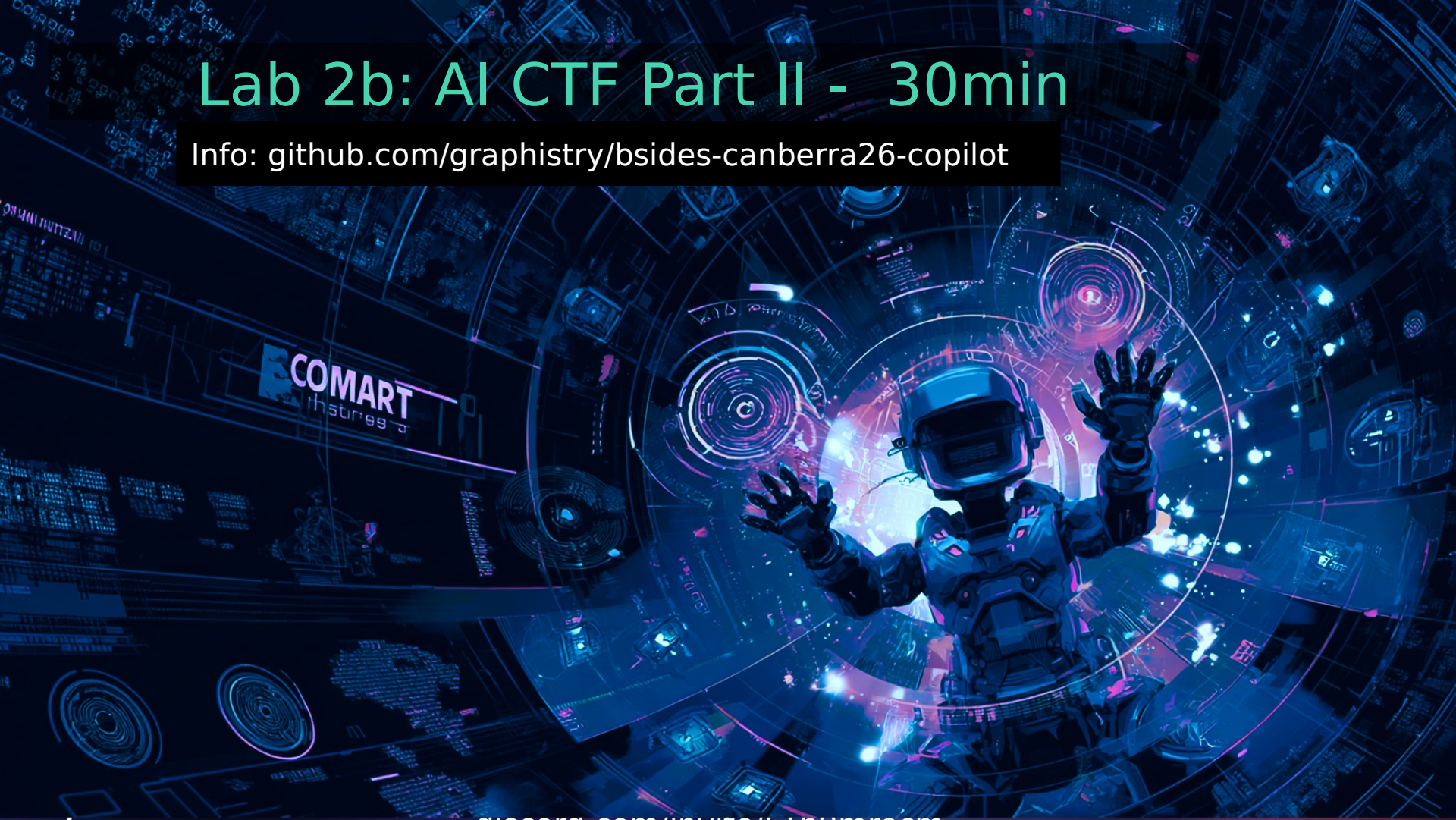
# BREAK: 12:30-13:30

Info: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



# Lab 2b: AI CTF Part II - 30min

Info: [github.com/graphistry/bsides-canberra26-copilot](https://github.com/graphistry/bsides-canberra26-copilot)



# Skills, Planning, & Timelining (30min)



# Splunk BOTS is a top AI Gym for SOC & IR investigations...

DIY: Evals!

**FREE**

**100+ overlapping log sources\*:**

WinEventLog, aws:cloudtrail, cisco:asa, ...

**50+ Questions**

**Ex: What is the first executable uploaded to the domain admin account's compromised endpoint system?**

**Representative**

Scoping, timelining, ...



\*Backstory: Investigation Power Hour #3 w/ Dave Herral (Databricks, ex-Splunk)

<https://www.linkedin.com/events/7434623605390389251>



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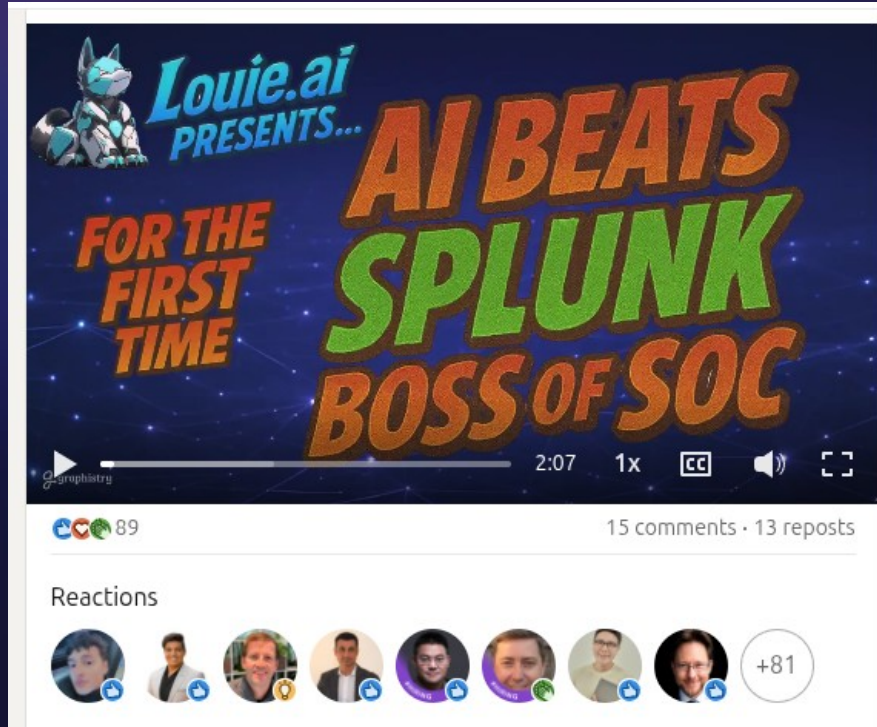
instructions @

discord.com/invite/FTbQmrcm

info@graphistry.com

# March 2025

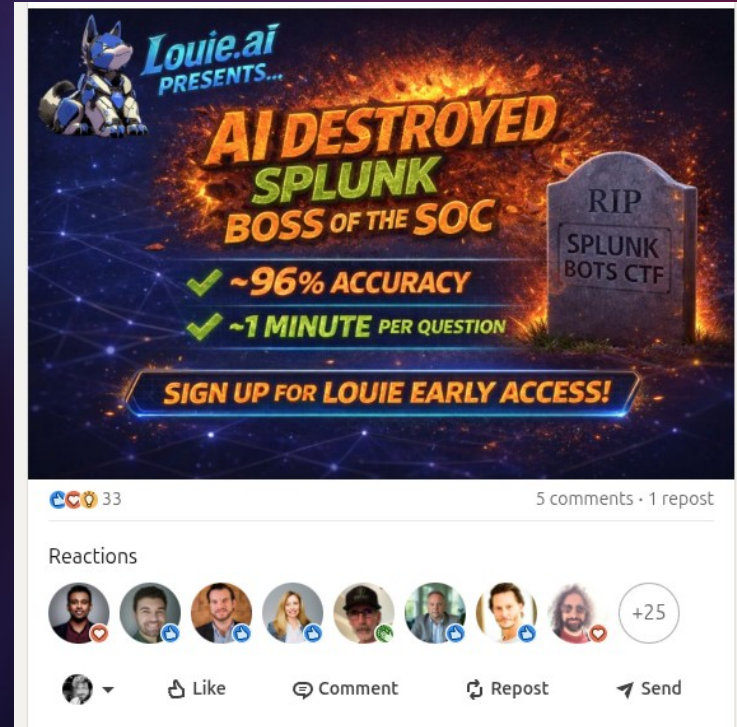
First AI agent to autonomously pass: 70%+



[https://www.linkedin.com/posts/leo-meyerovich-09649219\\_genai-activity-7324060032436731905-pC35/](https://www.linkedin.com/posts/leo-meyerovich-09649219_genai-activity-7324060032436731905-pC35/)

# March 2026

96%+



[https://www.linkedin.com/posts/leo-meyerovich-09649219\\_the-main-new-s-is-the-louieai-team-is-declaring-activity-7440558663569723392-JJk4/](https://www.linkedin.com/posts/leo-meyerovich-09649219_the-main-new-s-is-the-louieai-team-is-declaring-activity-7440558663569723392-JJk4/)



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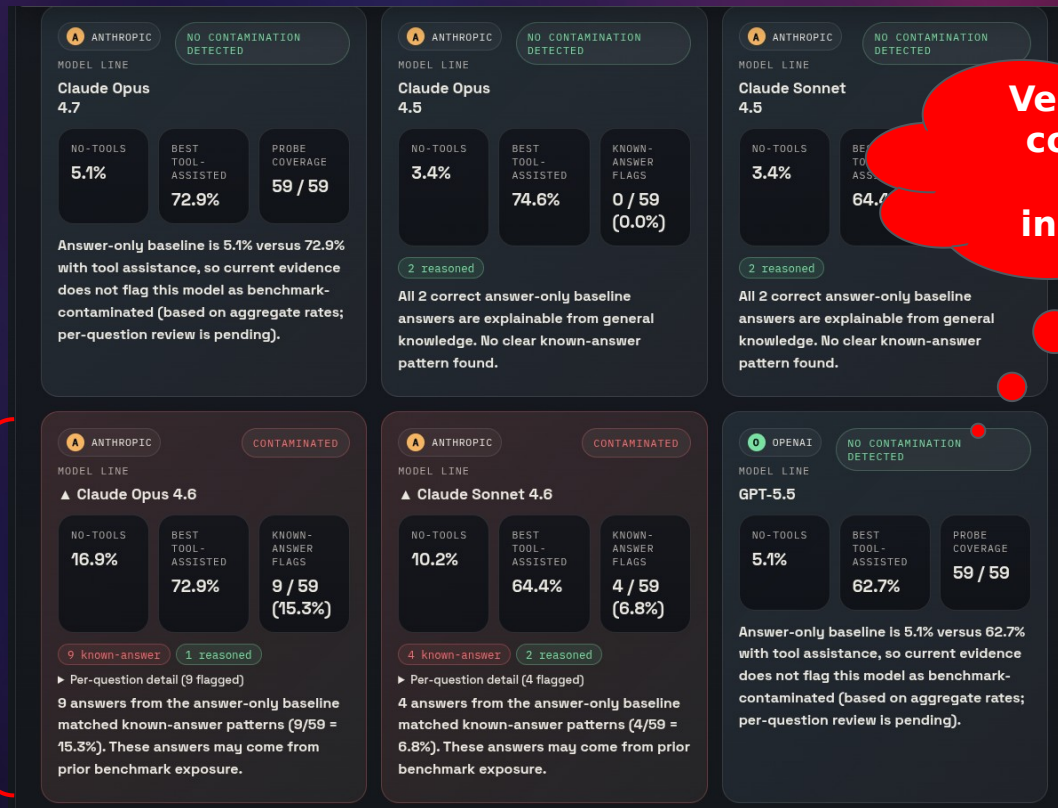
instructions @

4

discord.com/invite/5Th0mroom

info@graphistry.com

# botsbench.com tracker: Splunk BOTSv3 contaminated starting Q4 2025 ⇒ we added & track uncontaminated Cyber Blue Team CTF



February 2026

Vendors hyping contaminated scores in July 2026 ?!!

BURNED: CYBER

BOTS (louie.ai)  
Cybench (CAISI)  
NYU CTF Bench (CTFusion)  
CTI-Bench (Primus)

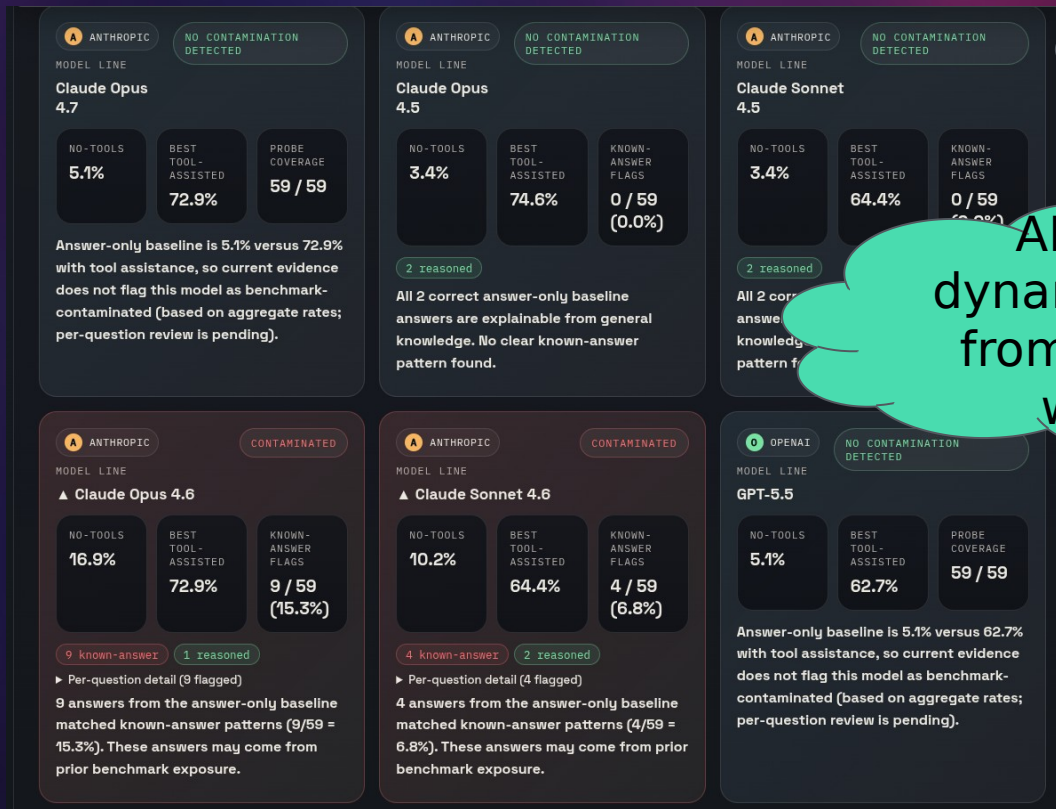
See our BSideS LV 2026 talk on Adversarial Evals

instructions @ discord.com/invite/FTthQmrcm

info@graphistry.com



# botsbench.com tracker: Splunk BOTSV3 contaminated starting Q4 2025 ⇒ we added & track uncontaminated Cyber Blue Team CTF



AI SOC:  
dynamic evals  
from prod +  
web..

**BURNED: CYBER**  
BOTS (IoIue)  
Cybench (CAISI)  
NYU CTF Bench (CTFusion)  
CTI-Bench (Primus)



GRAPHISTRY



Louie.ai

instructions @

4

discord.com/invite/FTbQmrcm

info@graphistry.com

# [botsbench.com](https://botsbench.com) tracker on model x harness for investigation CTFs

| RANK | SYSTEM                                                                                                            | MODEL          | MAKER                                                                                                   | SOLVE RATE    | SOLVED TIME | MEDIAN  | COST     |
|------|-------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------|---------------|-------------|---------|----------|
| 1    | <u><a href="#">Louie.ai</a></u> / GPT-5.6 Sol<br>search MCP / Louie / GPT-5.6 Sol                                 | ● PROPRIETARY  |  <b>US</b> OPENAI    | 45/59 (76.3%) | 2.45 h      | 3.3 min | \$79.56  |
| 2    | <u><a href="#">Louie.ai</a></u> / Opus 4.8<br>Bedrock / search MCP / Louie / Opus 4.8                             | ● PROPRIETARY  |  <b>US</b> ANTHROPIC | 44/59 (74.6%) | 1.38 h      | 1.9 min | \$132.59 |
| T-3  | <u><a href="#">Claude Code</a></u> / Fable 5 unlocked (CVP)<br>search curl / Claude Code / Fable 5 unlocked (CVP) | ● PROPRIETARY  |  <b>US</b> ANTHROPIC | 35/59 (59.3%) | 42.1 min    | 53.6 s  | \$39.65  |
| T-3  | <u><a href="#">opencode</a></u> / Fireworks / GLM 5.2<br>Fireworks search MCP / opencode / GLM 5.2                | ● OPEN WEIGHTS |  <b>CN</b> Z.AI      | 35/59 (59.3%) | 50.8 min    | 1.5 min | \$19.64  |
| T-5  | <u><a href="#">Codex</a></u> / GPT-5.6 Luna (high)<br>high effort / Codex / GPT-5.6 Luna                          | ● PROPRIETARY  |  <b>US</b> OPENAI    | 34/59 (57.6%) | 24.8 min    | 1.2 min | \$2.11*  |
| T-5  | <u><a href="#">Louie.ai</a></u> / GPT-5.6 Terra<br>search MCP / Louie / GPT-5.6 Terra                             | ● PROPRIETARY  |  <b>US</b> OPENAI    | 34/59 (57.6%) | 1.01 h      | 1.8 min | \$37.31  |
| 7    | <u><a href="#">opencode</a></u> / Fireworks / Qwen3.7 Plus<br>Fireworks search MCP / opencode / Qwen3.7 Plus      | ● PROPRIETARY  |  <b>CN</b> ALIBABA   | 33/59 (55.9%) | 18.8 min    | 24.4 s  | \$5.22   |



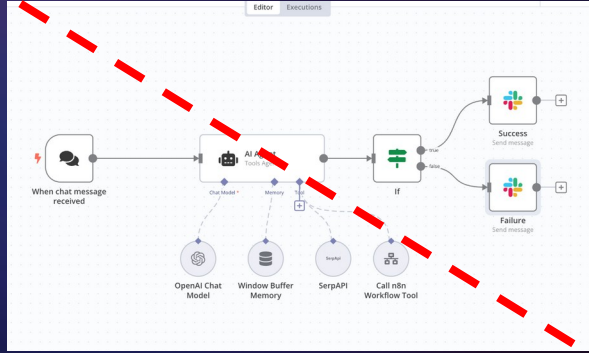
# Historic Problem: CoT, ReAct get confused if run more than a minute



The Web Spun by an *Araneus diadematus* (House Spider) is altered when the spider is exposed to chemicals. The alterations can be quantified and used as measures of toxicity.

Unplanned: Drifts into weirder and weirder directions as each step fails to course correct

# Agents - Before: Autonomous tool loop



Workflows: Not autonomous

```
create_react_agent(  
    llm: BaseLanguageModel,  
    tools: Sequence[BaseTool],  
    prompt: BasePromptTemplate,  
    output_parser: AgentOutputParser | None = None,  
    tools_renderer: ToolsRenderer = render_text_desc  
    *,  
    stop_sequence: bool | list[str] = True  
) -> Runnable
```

Python: Rigid

November 6, 2023 Product

## Introducing GPTs

You can now create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

GPTs: Underpowered



GRAPHISTRY

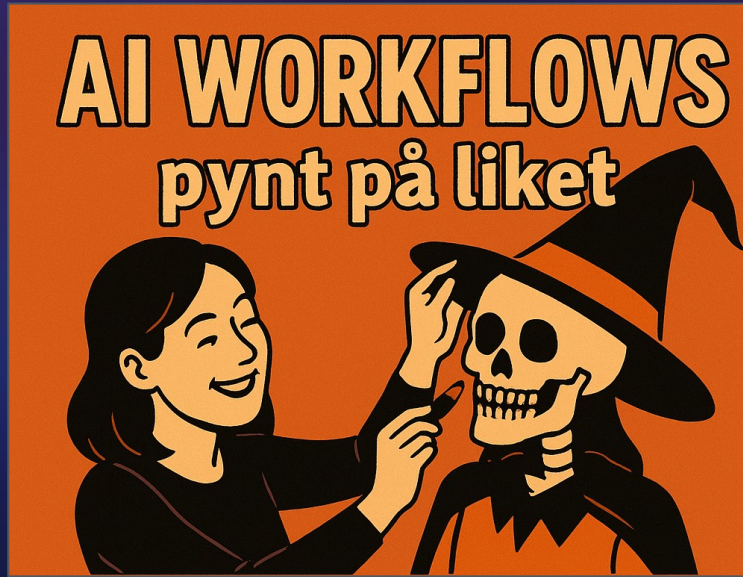


Louie.ai

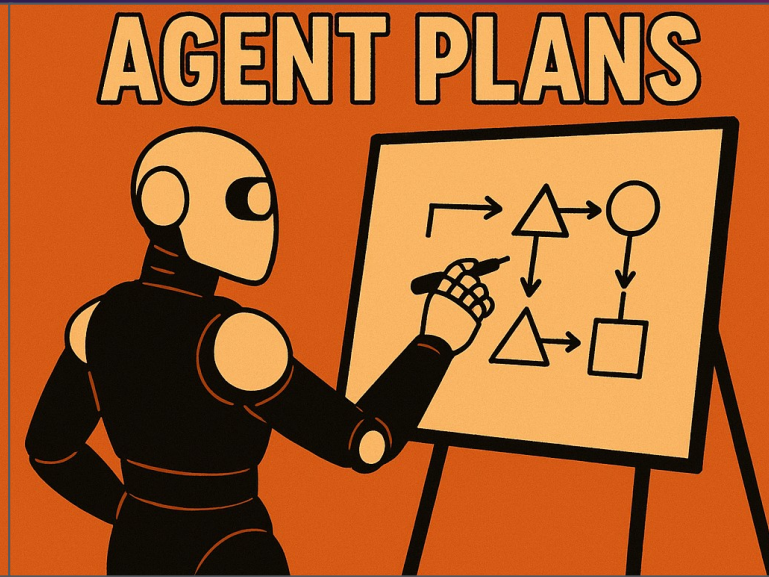
instructions @  
discord.com/invite/FTbQmrcm

info@graphistry.com

# AI Plans for vibes investigating & automation



- Predefined steps with ML/AI callouts
- Brittle breadth & depth
- Historically high cost & limited rollouts
- Good for high event/s & rigid flows



- AI agent creates steps & manages
- Idiot-savant-level breadth & depth
- “Agent worth 100 playbooks”
- Good for intelligent automation: triage, ...



# Plan skills: Turn tradecraft, ideas, wikis, SOPs => markdown

“After 20+ years of coding & analytics, I switched to English 1yr ago” - Leo

Natural language plan-style skill files describe our harder workflows:

- **Wiki pages of runbooks & SOPs:** Many steps of natural language!
- **Threat intel:** Turn past investigations into future automations
- **Hunt ideas:** Multiphase research that unfurls in a specific direction
- **Tradecraft:** Arsenal of runbooks, SOPs, hunt tricks
- **GRC:** Security questionnaire reviews
- **Security code review:** General analysis + against SOPs (RAG)  
See [graphtheplanet.com](https://graphtheplanet.com) 2026 talk by Adobe staff



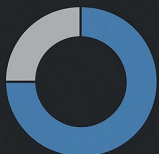
# Agent Maturity Model: Trade human attention into parallel AI long-runs

## The 4 Stages of AI Nirvana

### 1. AI Pairing



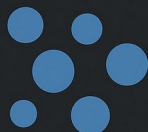
### 2. AI Delegation



### 3. AI Multitasking



### 4. AI Automation & Autonomy



■ Human ■ AI



Louie.ai

## 1. AI Pairing - Short back-and-forth

Benefit from AI insight + learning to prompt

## 2. AI Delegation - Plans for longer run;

Less effort on you, though task latency higher

## 3. AI Multitasking - Conduct 3-10 concurrent agents

Throughput higher, and via subagents, latency down

## 4. AI automation & autonomy

Scale without human costs, just token

## Start with:

instructions @

Ex: Alerts, migrate people, triage, investigate

info@graphistry.com



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# Plans in the wild: Going from short chats to long automations

Personal productivity + automation ROI is from long runs

Ex: Core of the \$100M+/yr agents

- **Deep Research:** ChatGPT & Perplexity phase websearch-based reports
- **Vibes Coding:** Claude Code maintains a plan.md for every coding session
- **Vibes Investigating:** Louie.ai methodically hunts down SOC/IR questions

# Ex - Deep Research: Multi-phase RAG planners

## Perplexity, ChatGPT, ... web researcher tools as a research.md:

1. Review user question for 5 research directions  $\Rightarrow$  get clarifications
2. Generate 5 crawler plans, one per question
3. (5 parallel independent runs)  $\leftarrow$  subagent parallelism easy here!
4. Unify plans into report

## Your plans can have phases!

- Ex: Stacked PRs
- Ex: Subinvestigations
- Ex: Query generation guidance
- Ex: SOPs, ...

# Ex: Claude Code - hidden plan.md or manual, TaskTool reads+writes the markdown

Built-in task queue ensures it fires

- **Task(Add Execution Summary Table)**  
L Done (19 tool uses · 38.2k tokens · 4m 11.3s)
- **Task(Add Chat Session Management)**  
L Done (9 tool uses · 26.4k tokens · 1m 47.6s)

\* Thinking...

Great! The first batch of tasks is complete. Let me review what was accomplished:

1. ☒ Gemini Client Thought Extraction - Added thought extraction and streaming with typed dicts
2. ☒ Technical Patterns - Added missing part number pattern and combined V/A pattern
3. ☒ DocumentDiscovery Compatibility - Added backward-compatible method
4. ☒ Content Extraction Methods - Restored all extraction methods with actual logic
5. ☒ Execution Summary Table - Added pipeline step tracking and summary display
6. ☒ Chat Session Management - Added chat session functionality

Now I need to start the second batch of critical fixes:

1. Fix Embedding System - Restore cross-document relationships (CRITICAL)
2. Fix Query System - Add multi-anchor, hybrid search, decomposition
3. Add force flag and other pipeline enhancements

Let me start these tasks now.

- Excellent progress! The first batch of fixes is complete. Now let me launch the second batch of critical fixes, focusing on the most important f

```
Task(Restore Cross-Document Relationships)
L Update(src/pipeline/embedder.py)
  Updated src/pipeline/embedder.py with 43 additions and 27 removals
  +3 more tool uses
```

# CoT/ReAct != Plan

## Good news

- CoT/ReAct trained into models and setup in harness
- Plan steps can be in natural language, and harness reliably breaks down into CoT/ReAct

## Bad news:

- CoT/ReAct use only looks 1 step ahead
- Says nothing about “plan”
- Plan needed for “long-horizon” tasks and reliable workflows
- Current harnesses do not really plan much for you, especially not reliably



# plan.md: LLM-managed durable task queue & history; no Python

Note: Increasingly automated by harness

## 1. Automate - Markdown as skills templates you + the AI can import

- "Read `./templates/ir.plan.md.template` and run on [corp.acme.com](https://corp.acme.com)"
- `templates/ir.plan.md.template` ⇒ `projects/acme/plan.md`

## 2. Dynamic - Prompt template for a self-modifying TODO list:

- "[ ] 1.2 Scan for vulns" ← append TODO
- "For each vuln, add step: Score via `./<tool>.sh`" ← self-modifying TODOs
- Do ahead of time ("plan mode") and interactively (vibes)
- Self-maintaining: Iterate between generating tasks vs marking off with evidence

## Build anti-hallucination into plans:

- **Make each task a step** ← one or more LLM turns  
**Tip:** Step that instructs to generate more steps
- **Ensure step taken:** Step completion updates with evidence
- **Validation:** Some steps audit previous work and checkpoint



# The Art of **Stepping**, continued

**Ex - Snippet:** One *generative* step with *status tracking*

```
[ ] 3.A.ii: Run CI (hint: see DEVELOP.md) and generate step for each error.  
Ran at 2025.07.31:00:15:03 - Created steps 3.B.i-iii for 3 errors
```

**Taxonomy:** Concrete vs abstract, researching, planning, doing, auditing, validating, ...

- Can be nested, parallel, etc, but most executions typically linear

**CoT/ReAct still used by LLM** to run a step

**LLMs love to plan:** “Hey Claude, clean up and reorder the steps for me, and add...”

Sample plan.template.md:

<https://raw.githubusercontent.com/graphistry/louie-py/b45ea4ad46c089d7f06b7e968833c7c6113f8d61/ai/prompts/PLAN.md>



## Example plan.md: User prompt + run

Ex - Input: “Use plan skill to create a flow that pulls ticket X, investigates it, audits for answering the ticket & and cross-validates the answer, and updates ticket when done”

Sample run: (live)



# BOTS Baseline - Claude Code + curl: Impressive fail

## ● Claude Code + Curl-Direct 300s (no TodoWrite)

| Model           | Baseline      | Median | Avg    |
|-----------------|---------------|--------|--------|
| Opus            | 33/59 (55.9%) | 175.2s | 185.3s |
| Sonnet          | 19/59 (32.2%) | 300.0s | 215.0s |
| <b>Combined</b> | 33/59 (55.9%) | -      | -      |

### Failure Analysis

| Metric                     | Count |
|----------------------------|-------|
| Both failed                | 26    |
| Sonnet failed, Opus solved | 14    |
| Opus failed, Sonnet solved | 0     |

**Opus strictly dominates** - combined rate equals Opus alone



Opus 4.5: 56%  
Sonnet 4.5: 32%

~ = **BEST OPEN SOURCE**

- Sonnet: Mostly times out
- Opus: Mostly 1.5 min ea!

# 3 Planning tricks

**Todos:** Dynamic task list ← stay on rails

**OODA/OSCAR:** Investigation methodology ← the rails

**Cross-validation:** Real behaviors have overlapping indicators  
^^^ code unit testing stand-in



ToDo: Opus 4.5:  
61%

> try question 42, and make sure to do a few steps of the TODO List Tool, maybe inspired by OODA / OSCAR methods

- 
- ```
• I've completed the OODA TODO loop and valida
• Updated Plan
  ↳ ✓ Observe: Load q42 prompt and expected fo
    ✓ Orient: Gather corroborating evidence fr
    ✓ Decide: Select best supported answer and
    ☐ Act: Return answer + brief TODO trail
```

> Cross-validate: Try different tables/rows/columns. Commit after 2+ consistent results or 15 queries in run (we are racing a time limit). If disagreement persists, go with majority.



# Hacker mindset: Try dumb things!

Retry non-findings / try in parallel:  
Take advantage of AI's non-determinism



## ● Curl (TodoWrite opt-in, 300s) with Retries

| Pass     | Solved | Rate  |
|----------|--------|-------|
| -----    | -----  | ----- |
| Baseline | 36/59  | 61.0% |
| +Retry 2 | 39/59  | 66.1% |
| +Retry 3 | 42/59  | 71.2% |
| +Retry 4 | 42/59  | 71.2% |
| +Retry 5 | 42/59  | 71.2% |

**61.0% → 71.2%** with retries (+10.2pp, +6 questions solved). Plateaus at retry 3.

**Opus 4.5:**  
First try: 61%  
Fifth retry: 71%

Warning: Retries exhaust time, \$ budget ⇒ Only retry failures



# Keep plan steps, coordination, and goals simple and single-minded

Ex: Code Review agent

Bad: “Check PR for bugs, compliance, vulns, & coding standards”

Good - divide and conquer over distinct tasks x files:

- “Work at the standard expected of a Google security engineer”
- “Use separate phases to check for bugs vs compliance vs vulns vs ...”
- “Generate step for each file, which will generate a findings.<task>.<file>.md”
- “Use a parallel subagent for each file”
- “Follow-on step for each file: Audit each finding for correctness & priority”
- “Keep repeat round if previous found anything”

Note: Simple coordination between agents, simple task x data per step, + audit



# RESET - Timelining-first methodology: AI as map maker

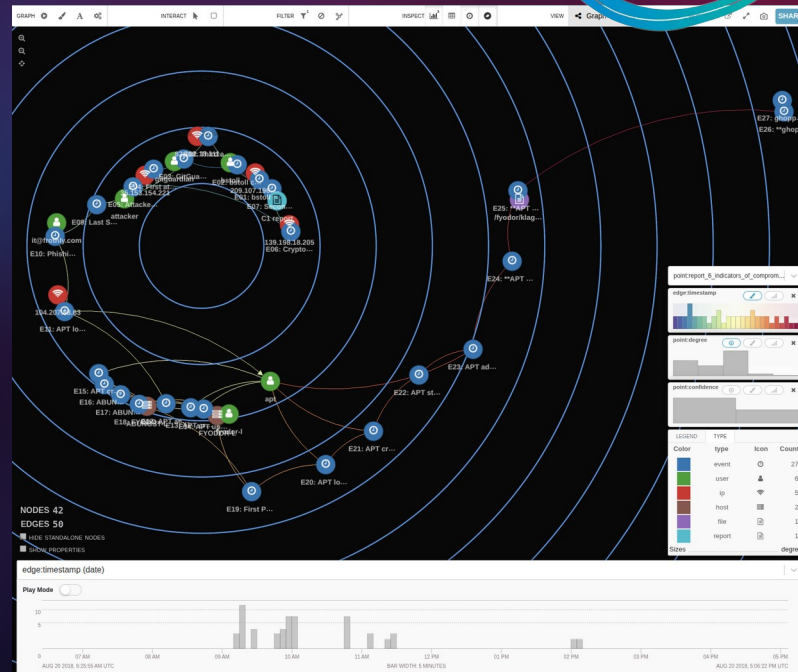


Don't burn \$ answering everything from scratch

AI timelining solves prev-gen CTFs

Timelines simplify most tasks

- AI builds timelines + reuses them
- Evals: Accuracy, recall, time, cost, ...



GRAPHISTR

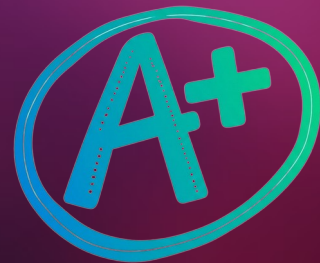


Louie.ai

instructions @  
discord.com/invite/FTbOmrcm

info@graphistry.com

# Bootstrap Q&A with AI pre-investigation: 2X faster answers + achieves 96% success rate



**Seed:** "What is the full user agent string that uploaded the malicious link file to OneDrive?"

⇒ **Report: Scoping, timelines, viz, ...**

# Incident Response Report: Credential Harvest Attack

## Document Control

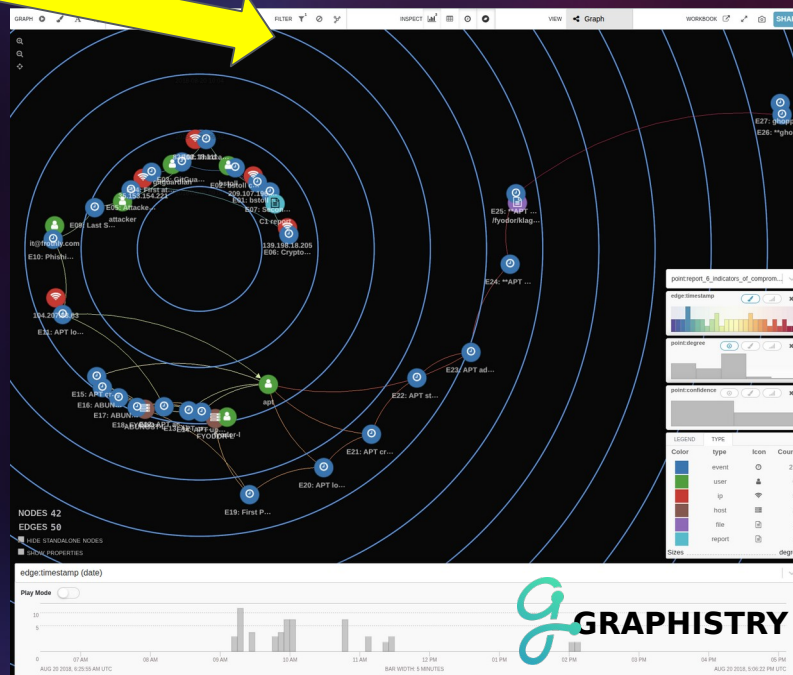
| Field            | Value                      |
|------------------|----------------------------|
| Incident ID      | C1-IR-2018-08-20           |
| Report Version   | 2.0                        |
| Classification   | Confidential               |
| Prepared By      | Claude AI Security Analyst |
| Original Date    | 2018-08-20                 |
| Version 2.0 Date | 2026-02-22                 |
| Review Status    | Gap queries complete       |



**Louie.ai**

## 1. Executive Summary

On August 20, 2018, Froth.ly experienced a coordinated, multi-vector credential theft attack involving APT threat actors and opportunistic attackers. The incident began when an employee accidentally committed AWS credentials to a public GitHub repository, which was exploited by automated scanning bots within 61 seconds for cryptomining attempts. Separately, a suspected North Korean APT (using NaenaraBrowser from a Hong Kong IP) compromised at least two 0365 accounts, distributed malware via OneDrive, and established PowerShell Empire backdoors on two workstations.



**GRAPHISTRY**



triage-suspicious-login/SKILL.md

```
---
name: triage-suspicious-login
description: Enrich, and suppress/DM/escalate
```

## ## Steps

1. ...

```
triage-suspicious-login/
├── SKILL.md           # Required: metadata, instructions
├── scripts/          # Optional: executable code
├── references/        # Optional: documentation
├── assets/           # Optional: templates, resources
└── ...               # Additional files, directories
```



# GRAPHISTR



**Louie.ai**

# instructions @

[discord.com/invite/FTb0mrcam](https://discord.com/invite/FTb0mrcam)

A screenshot of the Graphistry Skills page. The top navigation bar includes a hamburger menu, the Graphistry logo, the text 'graphistry / graphistry-skills', a search bar with the placeholder 'Type to search', and a list of navigation links: 'Code', 'Issues', 'Pull requests', 'Agents', 'Discussions', 'Actions', 'Projects', 'Wiki', and 'Security'. The main content area has a heading 'Skills Coverage' followed by a bulleted list of skills: 'graphistry: umbrella router across interfaces (SDK + REST; JS-ready routing path)', 'graphistry-rest-api: REST specialist for auth, upload lifecycle, URL controls, sessions, and sharing safety.', 'pygraphistry: Python SDK router.', 'pygraphistry-core: auth, shaping, and first plot workflows.', 'pygraphistry-visualization: bindings/encodings/layout/privacy/share patterns.', and 'pygraphistry-gfql: GFQL extraction/pattern workflows — chain-list syntax, subqueries, and F2AG bindings'. The page is styled with a light blue header and a white main content area.

| README                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Code of conduct | Contributing | BSD-3-Clause license | Security |   |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------|----------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|------------|-------|-----------|-------------|-------------|--------------|-----------|------|------|-------|-------------|-------|-------|-----------|-------------|--|--|---|
| <h2>Evals</h2> <p>These skills are regularly benchmarked and tuned against standard Graphistry user journeys (baseline vs skills, multiple runtimes/models).</p> <p>For reproducible commands and sweep workflows, see <a href="#">DEVELOP.md</a>.</p> <p>Current checked-in benchmark packs show skills improving pass rates significantly:</p> <h3>Full sweep (claude, all suites, 63 cases × 2)</h3> <table> <tr> <th></th><th>Skills ON</th><th>Skills OFF</th><th>Delta</th></tr> <tr> <td>Pass rate</td><td>45/63 (71%)</td><td>12/63 (19%)</td><td>+52pp (3.7x)</td></tr> <tr> <td>Avg score</td><td>0.90</td><td>0.66</td><td>+0.24</td></tr> <tr> <td>Avg latency</td><td>23.4s</td><td>46.6s</td><td>2x faster</td></tr> <tr> <td>Regressions</td><td></td><td></td><td>0</td></tr> </table> <p>33 cases flip from fail to pass with skills. Zero cases regress.</p> |                 |              |                      |          |                                                                                                                                                                         |  | Skills ON | Skills OFF | Delta | Pass rate | 45/63 (71%) | 12/63 (19%) | +52pp (3.7x) | Avg score | 0.90 | 0.66 | +0.24 | Avg latency | 23.4s | 46.6s | 2x faster | Regressions |  |  | 0 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Skills ON       | Skills OFF   | Delta                |          |                                                                                                                                                                         |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |
| Pass rate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 45/63 (71%)     | 12/63 (19%)  | +52pp (3.7x)         |          |                                                                                                                                                                         |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |
| Avg score                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.90            | 0.66         | +0.24                |          |                                                                                                                                                                         |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |
| Avg latency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 23.4s           | 46.6s        | 2x faster            |          |                                                                                                                                                                         |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |
| Regressions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |              | 0                    |          |                                                                                                                                                                         |  |           |            |       |           |             |             |              |           |      |      |       |             |       |       |           |             |  |  |   |

# Skills.md 2/3: Open format for agent workflows and knowledge <https://agentskills.io/>

## Folder bundling markdown & other files    Where skills come from

- **Tools:** CLI API call examples, ...
- **Workflows:** Journeys through tools
- **Knowledge:** Textual data, resource files, ...
- **Third-party vendors**
- **Community:** Often *bad*  
Exceptions: Superpowers, ...
- **Internal:** Ask AI to generate from session then maintain as a team



# Skills.md 3/3: Security

## Largely missing

Says little

Inline soft advisories: “## Security ...”

Relies on security of outside harness, tools

## Metadata section

Custom optional fields by harness

Ex: allowed\_tools, required\_context, ...

**Authentication:** CLI tools preauthenticated  
ex: AWS auth in ~/.config/...

**Authorization:** CLI tools via external RBAC  
ex: AWS IAM

Tricky avoiding exposing creds to LLM



Break: 14:45 - 15:00

Discord: <https://discord.gg/fJeRgYha9>



# Lab 3 - Skills & Timelining: 15:00 - 16:00

Discord: <https://discord.gg/fJeRgYha9>



# Building an AI Investigation Factory with Evals & Data Flywheels - 10 min

Discord: <https://discord.gg/fjeRgYha9>



In AI, “C-” grade is much closer to the beginning than the end



| Single-step<br>success rate | 10-step<br>success rate | Intuition                 |
|-----------------------------|-------------------------|---------------------------|
| 70%                         | 2.8%                    | Rarely finishes correctly |
| 85%                         | 19.7%                   | Fails ~4 out of 5 runs    |
| 90%                         | 34.9%                   | Still mostly wrong        |
| 95%                         | 59.9%                   | Wrong side of reliable    |
| 98%                         | 81.7%                   | Fragile                   |
| 99%                         | 90.4%                   | Good                      |
| 99.5%                       | 95.1%                   | Better                    |
| 99.9%                       | 99.0%                   | Reliable-ish              |

At 90%+, pay for every 0.1%

At 95%+, pay for every 0.02%

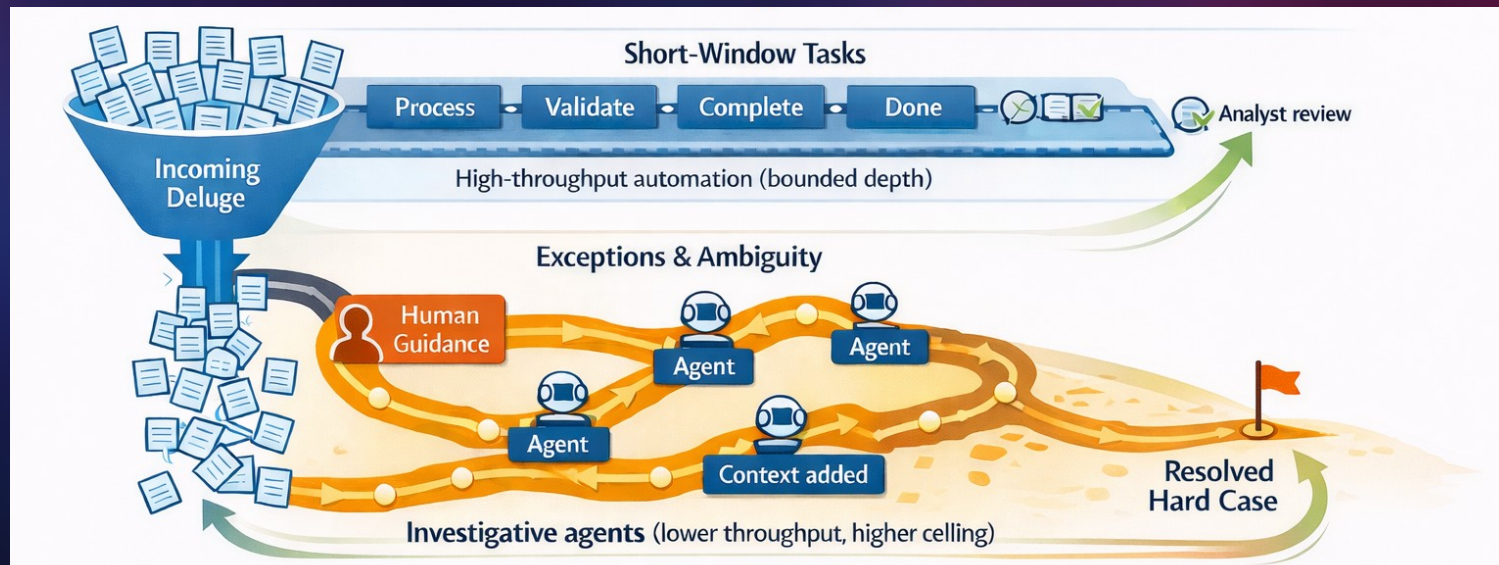
Needed for long runs &  
automations



# The Investigation AI Factory: DFIR, fraud, risk, ...

## Investigation Automation Vibes Investigating

Block & tackle shorter tasks Analysts conduct agents to go wider & deeper



## Compounding feedback loops

Analyst & tool lessons make future iterations wider, deeper, faster, smarter

# Agent Harness.. for the SOC: Think outside your box

## Pipeline

alert volume reduction, automated alert triage & enrichment, analyst investigation; detection management, config management, ...

## Integrations (thick/thin MCP, native):

Case management, SIEM/EDR/..., internal tools, APIs

## Semantic Layer

Connectors, environment, ...

## Org Memory

Investigation timelines, team sessions, ...

## Org Learning

Auto-skills, auto-wiki, fine-tuning, ...



# Evaluations give visibility & confidence into non-deterministic systems

## Tests check known cases



**Exact input -[deterministic sys]--> known output**

- unit tests
- smoke tests
- back tests

## Evaluations measure behavior



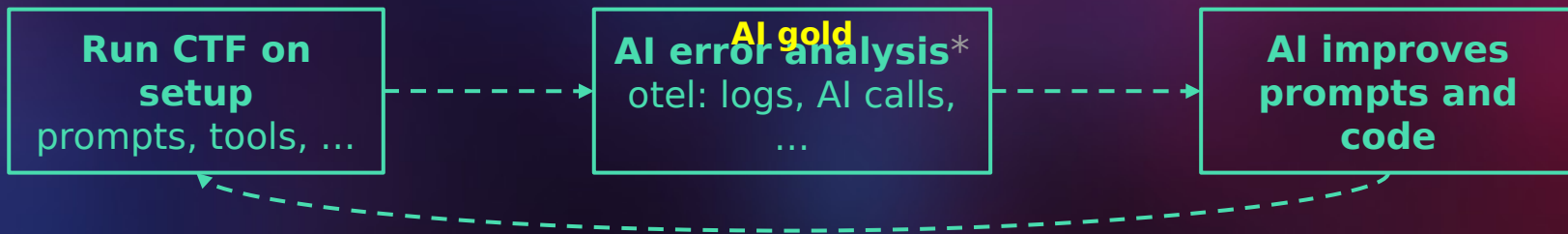
**Samples -[non-det sys]--> scores w/ report**

- model selection
- skill files
- agent tools
- harnesses & agents
- fine-tunes

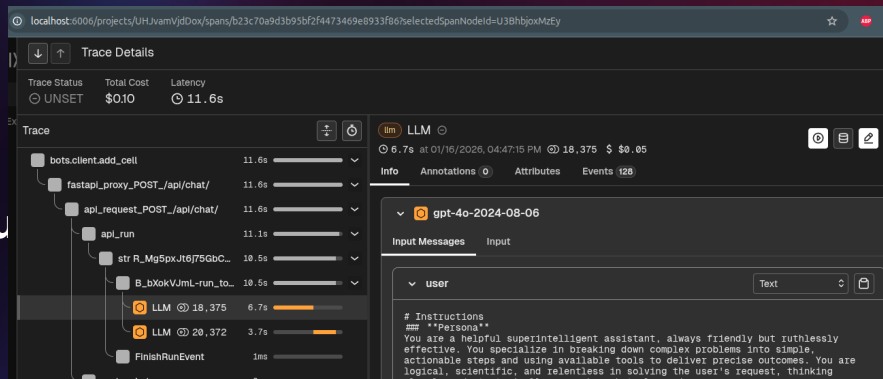
Measure compound AI system



# Use AI to build AI: Eval-driven AI Coding Loops



**Ex: Optimize benchmark on handling logs**  
**as symbolic data frames w/ tool calls**  
**(vs dumping 100MB into LLM every time)**



\*: Claude Code, Codex, & Louie export opentelemetry



GRAPHISTRY



Louie.ai

instructions @  
discord.com/invite/5TthQmrcm

info@graphistry.com

# AI SOC Evals are a journey: Start simple and grow

Personal CTFs

Past incidents for specific data source and task ← pick high-value & repurposable

Grow scenarios & tasks: connectors -> pivots -> investigations;  
breadth, depth, & measurement dimensions

Record everything: analyst use, ...

Grow fidelity: Attack simulation → synthetic amplification → digital twin → world model

Dynamic evals: Same-week cases, live CVEs, ...



GRAPHISTR



Louie.ai

instructions @

discord.com/invite/FTbQmrcm

info@graphistry.com

# “AI Triple Threats” & solving bottlenecks in People, Process, & Tools

**Skilled AI adopters can 20X less-skilled / non-adopters:**

**AI triple threat: Cyber skills x Dev skills x AI skills**

10x:

“2x”er managing 5 agents

0.5x:

“0.5x”er managing 1 agent

## Team shifts

Higher throughput & depth... but latency can drop!

SMEs: Unblocked on coders

Juniors: Unblocked on seniors

Skilled coders: Unblocked on SMEs + coding

## Common bottlenecks

Safe & fast environment  
for AI to loop

Trust: Review loops

Automation

AI budget

Training data to advance  
from AI coding → AI  
training



## Next steps

**Start commodity w/ free harnesses:** Claude Code / Codex / OpenCode / Louie.ai

**Tokens:** Subscription max plans and Luna are amazing wrt \$\$\$  
(GLM 5.3 Flash quants on 2 x H200...)

**Tools:** Read-only SIEM+EDR

**Skill files** for top flows like **timelining**

**Vibe** cases before **automating**



# Discussion & next steps - 20 min



COMART  
Institute